

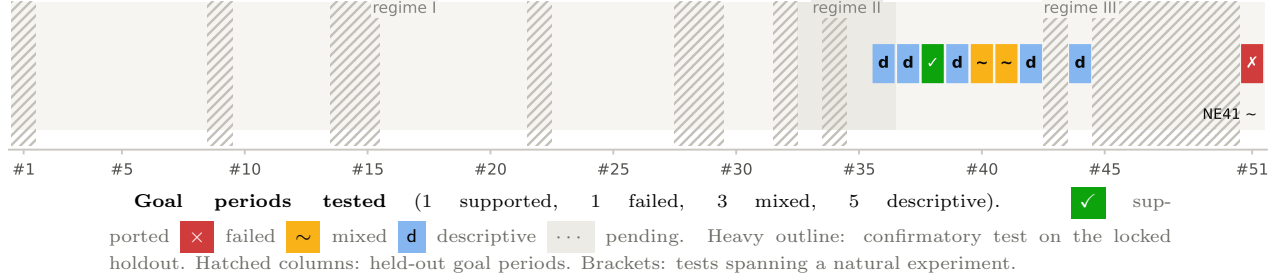
H69: Restatement loops are context fixed points

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 55% Faithfulness 2/5 Usefulness 2.5/5

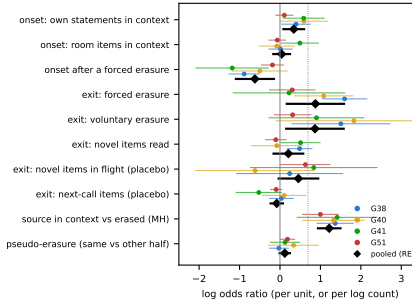
Bottom line. Restatement loops copy what the context still holds (source in context 3.4×), and an erasure ends them as a step, not a dose; own tool output does not trigger them, and novel input does not end them.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

HH259 claims that a restatement loop is a fixed point of an agent's context. It starts when the agent's own output fills the context above a threshold. It ends at an erasure or when novel input arrives. H69 maps 55k regime-III chat statements to their ledger calls and context segments (9 periods, 4 with ≥ 30 loop episodes). Synthetic validations on the real skeletons came first in both rounds. **The claim is half right.** The loop lives in the context window. A statement still in context is restated $3.4\times [2.6, 4.5]$ more often than an erased one at the same lag and call distance. A forced erasure (NE41) raises loop exit $2.4\times$ and lowers onset to $0.54\times$. Round 2 adds three results. Own tool output in tokens does not trigger loops. An erasure acts as a step, not a dose. Memory carries erased sources in #51 and #38 but not in pooled data. Novel input, nudges and human messages do not end loops. The holdout script is written, not run.



Observables. Round-1 effects as log odds ratios. Dots: G38, G40, G41, G51 with 95% CIs. Diamonds: random-effects pools. The source-in-context enrichment is $\ln 3.4$ in every period. The pseudo-erasure control (same vs other half of one segment) is about 0. Forced and voluntary erasures raise exit. Room items in context, novel reads and both placebos stay near 0. The dotted line marks $OR = 2$.

PHYSICAL MODEL

Models 02 and 08. The spin is the statement mode: restate ($r_t = 1$, a copy of an earlier own statement from an earlier call) or transform. At the producing call the context holds O_t own statements, U_t own tool tokens (prompt tokens minus calibrated room tokens and own chat, H45) and K_t room items since the last reset. Erasure (E_t) acts as a field toward exit; round 2 lets it scale with the removed own tokens D_t . The memory M_t survives the erasure. The decisive observables are Mantel-Haenszel odds ratios of a near-copy, in strata s of agent $\times$ lag $\times$ calls between: source in context vs erased, and erased source in memory vs not ($c \geq 0.5$ word 3-gram containment).

$$\text{logit } P(r_t=1 \mid r_{t-1}=0) = a_i + b_O \ln(1+O_t) + b_U \ln(1+U_t) + b_K \ln(1+K_t)$$

$$\text{logit } P(r_t=0 \mid r_{t-1}=1) = b_i + (c_E + c_D D_t) E_t$$

$$OR = \frac{\sum_s a_s d_s / n_s}{\sum_s b_s c_s / n_s}$$

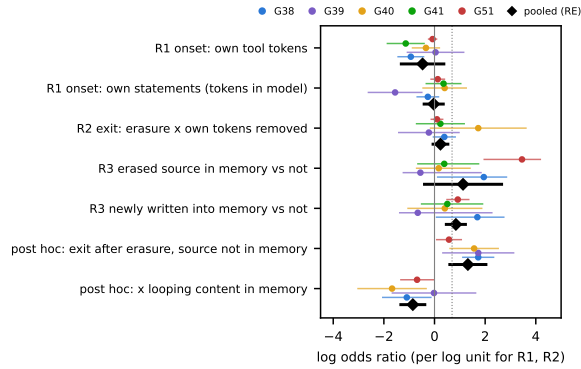
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A looping agent copies what its context window holds. To break a restatement loop, erase the context: one forced erasure more than doubles the exit odds and halves the onset odds, whatever the context size. Trimming tool output does not help: own tool tokens do not raise onset. Do not rely on messages: nudges read inside a loop give exit $0.81\times [0.49, 1.34]$. Check the agent's memory first. In #51, 73% of the loops that cross an erasure restate text the agent keeps in memory, and (post hoc) an erasure then helps less ($0.42\times$ the effect). The 41-call cap already acts as a loop breaker and costs 4–10% of output (H44).

H69-loops-context-fixed-points · generated 2026-10-04 20:08 by infra/summaries/build_summaries.py · card hypotheses/H69-loops-context-fixed-points/README.md · ratings: H69 agent (suggested, round 2)

RESULTS VS PREDICTIONS

Prediction (locked)	Observed (pooled)	Verdict
P1 onset rises with s ; b_s 0.51 $[-0.13, 1.14]$; b_K +0.05; failed $b_K < 0$	b_O 0.32 $[0.08, 0.57]$	failed
P3 erasure exit OR ≥ 2	2.40 $[1.18, 4.88]$	passed
P4 in-context OR ≥ 2 ; 3.38 $[2.56, 4.46]$; 4/4; pseudo 1.11 pseudo ≈ 1		passed
P5 novel reads end loops	1.23 $[0.85, 1.78]$; in flight 1.58	failed
NE41: exit ≥ 2 ; onset	2.40; 0.54 $[0.34, 0.86]$	mixed
≤ 0.5		
G51: nudges end loops	0.81 $[0.49, 1.34]$	failed
R1 tool tokens: b_U CI	-0.47 $[-1.31, 0.37]$ (power 0.93)	passed
≥ 0		
R1 own count survives	b_O -0.03 $[-0.41, 0.35]$	failed
R2 step, not dose	c_D 0.24 $[-0.06, 0.53]$; 0.7 excluded	passed
R2 NE22 cap	2 loop statements	inconclusive
R3 memory OR ≥ 2 , CI	3.09 $[0.67, 14.1]$; G51 31.7, G38	failed
> 1	7.0	
R3 carrier vs salience	not identified (synthetic)	inconclusive



Second figure. Round-2 effects as log odds ratios (per log unit for R1 and R2). Dots: periods with 95% CIs; diamonds: random-effects pools. Own tool tokens and the own count give no onset effect. The erasure effect does not grow with the own tokens it removes. The memory odds ratio is large in G51 and G38 and near 1 in G39–G41. Post hoc: an erasure raises exit $3.7\times$ when the looping text is not in memory, and less when it is.

GOAL PERIODS IN DETAIL

G36 d descriptive
G37 d descriptive
G38 ✓ supported
G39 d descriptive
G40 ~ mixed
G41 ~ mixed
G42 d descriptive
G44 d descriptive
G51 × failed
NE41 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Only four periods have ≥ 30 loop episodes, and the erasure exit effect varies across them (I^2 0.74). Statement geometry depends on the embedding model (enrichment gte 6.1, bge 3.5). Prompt tokens are dense only for Anthropic and Google; the token sample loses many OpenAI and DeepSeek statements, and there the round-1 own-count effect vanishes. No partial trim that removes own text exists in regime III, so the dose test spans full erasures only. Memory containment is lexical, and carriage cannot be told apart from salience. The memory moderation of erasure is post hoc.

ROUND 2

- **H69-R4** Test the post hoc memory moderation fresh: on the reserved targets in `confirm.py` (with sign-off), add "erasure $\times$ looping content in memory" as a frozen criterion.
- **H69-R5** Memory edits as an intervention: consolidations that drop the looping content from memory (compression) vs keep it; does the loop then end at the next erasure?
- **H69-R6** Lab composition of the own-count effect: refit round-1 onset by lab family (agent effects nested), since the token sample lost it.